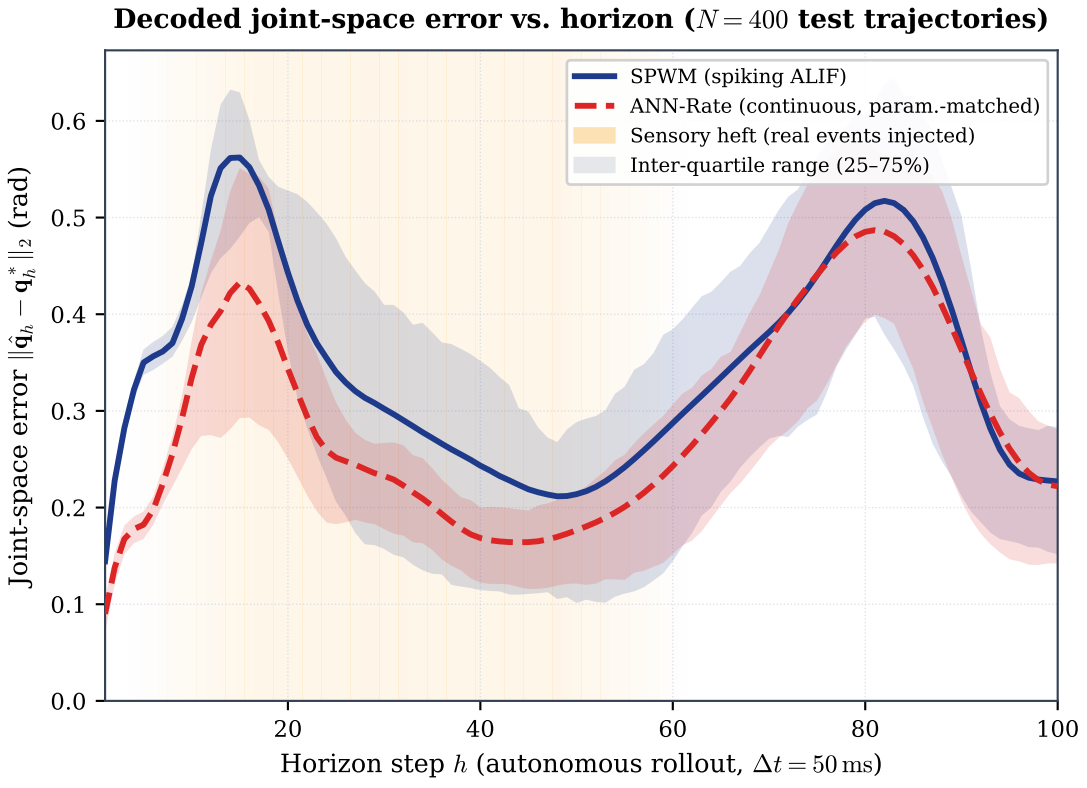
Decoded joint-space error vs. horizon ($N = 400$ test trajectories)
SPWM (spiking ALIF)
ANN-Rate (continuous, param.-matched)
Sensory heft (real events injected)
Inter-quartile range (25–75%)
Joint-space error $\|\hat{\mathbf{q}}_h - \mathbf{q}^*_h\|_2$ (rad)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
20
40
60
80
100
Horizon step h (autonomous rollout, $\Delta t = 50\,\mathrm{ms}$)